CURRICULUM VITAE

Daniel Arribas-Bel

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Education

2010 PhD Economics, Universidad de Zaragoza (Spain).
Essays on the Distribution and Spatial Structure of Cities.
Supervisor: Fernando Sanz-Gracia.
Committee: Prof. Michael J. Batty (chair), Prof. Sergio J. Rey, Prof. Enrique Lopez Bazo, Dr. Rafael Boix Domenech.

2008 MSc. Economics, Universidad de Zaragoza (Spain).
Kangaroos, Cities and Space: a First Approach to the Australian Urban System.

2006 BS. Economics, Universidad de Zaragoza (Spain).

Current Academic Appointments

2022- Professor in Geographic Data Science, Department of Geography and Planning, University of Liverpool.

2025- Co-Director, Imago Smart Data Research UK Data Service for Imagery.

Honorary Appointments

2022- Affiliate, Center for Open Geographical Science, San Diego State University.

2016- Fellow, Center for Spatial Data Science, University of Chicago.

2006- Researcher, ADETRE Research Group, Universidad de Zaragoza.

2020-2022 Honorary Senior Lecturer, Department of Computer Science, University of Liverpool.

2018-2022 Affiliate, Center for Geospatial Sciences, University of California Riverside.

2017-2019 Member, Smart Cities UB Chair, Universitat de Barcelona.

Editorial Appointments

2019- Co-editor, *Environment and Planning B - Urban Analytics and City Science*

2025- Editorial Board, *Annals of the American Association of Geographers*

2021- Editorial Board, *Journal of Geographical Systems*

2019- Editorial Board, *Spatial Economic Analysis*

2017- Editorial Board, *Geographical Analysis*

2020-2024 Associate Editor, *Journal of the Royal Statistical Society Series A - Statistics in Society*

2016-2022 Editorial Board, *Transactions in GIS*

2015-2018 Book Reviews Editor, *Environment and Planning B - Urban Analytics and City Science*

Other Appointments

2021- Steering committee, workstream coordinator Urban Analytics, The OECD Laboratory for Geospatial Analysis. OECD.

2016-2022 Programme Director, MSc in Geographic Data Science, University of Liverpool.

2017-2022 Chair, Quantitative Methods Research Group, Royal Geographical Society.

Prior Academic Appointments

2021-2025 Deputy Programme Director, Urban Analytics, The Alan Turing Institute.

2020-2023 ESRC Fellow, The Alan Turing Institute.

2018-2022 Senior Lecturer in Geographic Data Science, Department of Geography and Planning, University of Liverpool.

2015-2018 Lecturer in Geographic Data Science, Department of Geography and Planning, University of Liverpool.

2012-2016 Research Associate, GeoDa Center for Geospatial Analysis and Computation, Arizona State University.

2014-2015 Lecturer in Human Geography, School of Geography, Earth and Environmental Sciences, University of Birmingham (UK).

2012-2014 Postdoctoral researcher, Department of Spatial Economics, Vrije Universiteit. Amsterdam (The Netherlands).

2011-2012 Postdoctoral Research Associate, GeoDa Center for Geospatial Analysis and Computation, Arizona State University.

2006-2010 PhD Student, Department of Economic Analysis, Universidad de Zaragoza.

Honors and Awards

2026 *Media Achievement Award* for the GLaD podcast. American Association of Geographers. Joint with Rachel Franklin and Levi Wolf.

2024 *Taylor & Francis Award* for "excellence in the practice and promotion of teaching in higher education". Royal Geographical Society.

2021 *"Keeping the conversation going" - The use of Teams to sustain engagement in Geographic Data Science learning.* Faculty Learning, Teaching and Student Experience Award Winner. Faculty of Science and Engineering. University of Liverpool.

2020 *Spatial Data Scientist of the Year* (inaugural). Spatial Data Science Conference (Online). CARTO.

2006 *Premio extraordinario de licenciatura* (Best student of the year in economics at the University of Zaragoza).

Research Income (≈ *£12.5M, £7.7M as PI*)[1]

Awards

2025 *Smart Data Research UK* — "Embedding embeddings across social research and policy (EMBED2Social)". Opportunity Fund (additional funding to the Imago Data Service for Imagery). PI. 2025/26. £54,761

2024 *Economic and Social Research Council, UK Research and Innovation* — "The SDR UK Imagery Data Service for Sustainability, Prosperity and Wellbeing (Imago)". Smart Data Research UK Data Services. PI. 2024/29. £6,788,641

[1] Converted to GBP at ECB reference rates of 25 September 2026.

The Royal Society — "Learning from UK-France cross border railway transportation: border effect modelling and operational modes for the China-Laos Railway". International Exchanges. CoI. £12,000

European Space Agency — "EuroFab: European Urban Fabric Classification Using Artificial Intelligence". Tender. CoI. €250,000

European Space Agency — "EO Informed Agent Based Models for Digital Twins Applications". Tender. CoI. €399,739

2023 *Geospatial Commission* — "National Land Data Programme Turing Demonstrator Extension". Grant funding agreement with the Geospatial Commission (Department for Science, Innovation and Technology). PI. £99,955

2022 *Geospatial Commission* — "Using scenario-based land use modelling to identify the impacts from policy demands on land use change in a large conurbation". Grant funding agreement for the National Land Data Programme - Land Use Demonstrator. PI. 2022/23. £182,329

The Alan Turing Institute — "The Science of Cities and Regions". Out of budget process. CoI. 2022/25. £2,996,046

Economic and Social Research Council (*ESRC*) — "Supporting Post Pandemic Recovery and Resilience through New Forms of Data". ESRC AHRC UK South Korea SSH Connections Grants. CoI. 2022/23. £138,893.36

Fundación La Caixa — "Building and shaping cities for a healthy and prosperous future". Social Research Grants. CoI. 2022/23. €100,000

Engineering and Physical Sciences Research Council (*EPSRC*) — "ITINERANT: InequaliTies IN Experiencing uRbAn fuNcTion. Cuebiq data subscription". CoI. 2022. £14,000

2021 *Engineering and Physical Sciences Research Council* (*EPSRC*)/*The Alan Turing Institute* — "InequaliTies IN Experiencing uRbAn fuNcTion (ITINERANT)". PI. 2021/22. £96,624

Natural Environment Research Council (*NERC*) — "Nitrogen Fixation in the Arctic Ocean". Standard grant, `NE/T001240/1`. CoI. 2021/23. £733,137

2020 *Economic and Social Research Council* (*ESRC*) — "Learning an urban grammar from satellite data through AI". Joint ESRC/The Alan Turing Institute Fellowship, `ES/T005238/1`. PI. 2020/23. £426,224

Maritime Research and Innovation UK (*MarRI-UK*) — "Assuring safe port navigation by applying machine learning (ML) to wave data for automated monitoring of changes in nearshore bathymetry". CoI. 2020/21. £169,232

2019 *University of Liverpool Industrial Strategy Fund* — "Envisaging Liverpool's Future Digital City Region and Smart Governance". Together with Prof. Alex Singleton (PI) and Prof. Mark Boyle (CoI). CoI. £23,000

Kings College London Faculty Education Fund — "Conjuring: Self-Contained Jupyter Hub for Teaching". Together with Dr. Jon Reades (PI) and Dr. Steffen Zschaler (CoI). CoI. £4,975

2018 *ODA Research Seed Fund, University of Liverpool* — "Past and future of urban structures in Latin American cities". PI. £8,887

2017 *Santander Bank* — "Travel Grant". PI. £1,000

2014 *FOSTER EU-funded (FP7)* — "Workshop on Open Workflows". Together with Thomas de Graaff. CoI. €2,150

GeoDa Center for Geospatial Analysis and Computation, Arizona State University — "Space in Hierarchical Linear Models using Python". PI. $18,000

2007 *Spanish Ministry of Education and Science* — "University Staff Training Programme (in Spanish, "Formación de Personal Universitario", FPU)". Pre-Doctoral grant. 2007/10. €48,000

Projects (as researcher)

2009-2013 *Migrant Diversity and Regional Disparity in Europe*. Peter Nijkamp (PI) and Jacques Poot (Co-PI). Post-doctoral researcher (June, 2012-March, 2014). Sponsor: NORFACE research program on migration. €2,548,920.

2011-2013 *Affordable Housing and Walkable Neighborhoods A National Urban Analysis*. Julia Koschinsky and Emily Talen (PI and Co-PI). Researcher (January-May, 2012). Sponsor: U.S. Department of Housing and Urban Development. US$500,000.

Flexible Geospatial Visual Analytics and Simulation Technologies to Enhance Criminal Justice Decision Support Systems. Luc Anselin and Serge Rey (PI and Co-PI). Researcher (January-December, 2011). Sponsor: U.S. Department of Justice, National Institute of Justice. US$1.4 million.

2010 *Estrategia de Zaragoza y su entorno en el horizonte 2020*. Scholar. Sponsor: Ebrópolis.

2009-2012 *Aspectos teóricos y empíricos asociados al fenómeno de urban sprawl* `ECO2009-09332`. Scholar. Sponsor: Ministry of Science and Innovation, Government of Spain. €42,900.

2007-2009 *La distribución del tamaño de las ciudades: regularidades empíricas y modelos teóricos explicativos* `SEJ2006-04893`. Scholar. Sponsor: Ministry of Science and Education, Government of Spain. €34,485.

Publications

Books

2023 Rey, S.; Arribas-Bel, D.; Wolf, L. J. "Geographic Data Science with Python". Chapman & Hall/CRC. https://geographicdata.science/book/

Peer-reviewed journal articles

2026 Sato, Y.; Pietrostefani, E.; Mahabir, R.; Arribas-Bel, D. "City2Graph: A Python library for Heterogeneous Graph Neural Networks and spatial analysis in urban systems". *Computers, Environment and Urban Systems*, 130, 102492. 10.1016/j.compenvurbsys.2026.102492

Metzler, B.; Fleischmann, M.; Arribas-Bel, D. "Is a (satellite) image worth a thousand data points? Comparing machine learning approaches to predict environmental and social inequalities in England". *PLOS ONE*, 21(8), e0356472. 10.1371/journal.pone.0356472

2025 Murphy, J.; Mahabir, R.; Phinn, S.; Arribas-Bel, D. "Navigating coastal resilience: Exploring population density and coastal distance dynamics along the gold coast". *Global Challenges & Regional Science*, 3, 100019. 10.1016/j.gcrs.2025.100019

2024 Fleischmann, M.; Arribas-Bel, D. "Decoding (urban) form and function using spatially explicit deep learning". *Computers, Environment and Urban Systems*, 112, 102147. 10.1016/j.compenvurbsys.2024.102147

Kenyon, G.; Arribas-Bel, D.; Robinson, C. "Extracting Features from Satellite Imagery to Understand the Size and Scale of Housing Sub-Markets in Madrid". *Land*, 13, 575. 10.3390/land13050575

Kenyon, G.; Arribas-Bel, D.; Robinson, C. "Intra-urban house prices in Madrid following the financial crisis: an exploration of spatial inequality". *npj Urban Sustain*, 4, 26. 10.1038/s42949-024-00161-0

2023 Calafiore, A.; Samardzhiev, K.; Rowe, F.; Fleischmann, M.; Arribas-Bel, D. "Inequalities in experiencing urban functions. An exploration of human digital (geo-)footprints". *Environment and Planning B - Urban Analytics and City Science*. 10.1177/23998083231208507

Salat, H.; Carlino, D.; Benitez-Paez, F.; Zanchetta, A.; Arribas-Bel, D.; Birkin, M. "Synthetic population Catalyst: A micro-simulated population of England with circadian activities". *Environment and Planning B - Urban Analytics and City Science*. 10.1177/23998083231203066

Murphy, J.; Plater, A.; Bird, C.; Arribas-Bel, D. "Spatial-temporal variability: characterisation of a beach system using high resolution radar data". *Frontiers in Marine Science*, 10. 10.3389/fmars.2023.1142077

Owen, D.; Arribas-Bel, D.; Rowe, F. "Tracking the Transit Divide: A Multilevel Modelling Approach of Urban Inequalities and Train Ridership Disparities in Chicago". *Sustainability*, 15(11). 10.3390/su15118821

2022 Arribas-Bel, D.; Fleischmann, M. "Spatial Signatures - Understanding (urban) spaces through form and function". *Habitat International*, 128. 10.1016/j.habitatint.2022.102641

Fleischmann, M.; Arribas-Bel, D. "Geographical characterisation of British urban form and function using the spatial signatures framework". *Scientific Data*, 9, 546. 10.1038/s41597-022-01640-8

Rowe, F.; Calafiore, A.; Arribas-Bel, D.; Samardzhiev, K.; Fleischmann, M. "Urban exodus? Understanding human mobility in Britain during the COVID-19 pandemic using Meta-Facebook data". *Population, Space and Place*, 29, e2637. 10.1002/psp.2637

Patias, N.; Rowe, F.; Arribas-Bel, D. "Local urban attributes defining ethnically segregated areas across English cities: A multilevel approach". *Cities*, 132. 10.1016/j.cities.2022.103967

Samardzhiev, K.; Fleischmann, M.; Arribas-Bel, D.; Calafiore, A.; Rowe, F. "Functional signatures in Great Britain: A dataset". *Data in Brief*, 43. 10.1016/j.dib.2022.108335

Comber, S.; Park, S.; Arribas-Bel, D. "Dynamic-IMD (D-IMD): Introducing activity spaces to deprivation measurement in London, Birmingham and Liverpool". *Cities*, 127. 10.1016/j.cities.2022.103733

Chen, M.; Liu, Y.; Arribas-Bel, D.; Singleton, A. "Assessing the value of user-generated images of urban surroundings for house price estimation". *Landscape and Urban Planning*, 226. 10.1016/j.landurbplan.2022.104486

Singleton, A.; Arribas-Bel, D.; Murray, J.; Fleischmann, M. "Estimating generalized measures of local neighbourhood context from multispectral satellite images using a convolutional neural network". *Computers, Environment and Urban Systems*, 95. 10.1016/j.compenvurbsys.2022.101802

2021 Patias, N.; Rowe, F.; Arribas-Bel, D. "Trajectories of neighbourhood inequality in Britain: Unpacking inter-regional socioeconomic imbalances, 1971-2011". *The Geographical Journal*. 10.1111/geoj.12420

Rey, S. J.; Anselin, L.; Amaral, P.; Arribas-Bel, D.; Cortes, R. X.; Gaboardi, J. D.; Kang, W.; Knaap, E.; Li, Z.; Lumnitz, S.; Oshan, T. M.; Shao, H.; Wolf, L. J. "The PySAL Ecosystem: Philosophy and Implementation". *Geographical Analysis*. 10.1111/gean.12276

Arribas-Bel, D.; Green, M.; Rowe, F.; Singleton, A. "Open data products - A framework for creating valuable analysis ready data". *Journal of Geographical Systems*. 10.1007/s10109-021-00363-5

Wolf, L. J.; Anselin, L.; Arribas-Bel, D.; Mobley, L. R. "On Spatial and Platial Dependence: Examining Shrinkage in Spatially Dependent Multilevel Models". *Annals of the American Association of Geographers*. 10.1080/24694452.2020.1841602

Alharbi, Y.; Arribas-Bel, D.; Coenen, F. "Sustainable Development Goal Relational Modelling and Prediction: Introducing the SDG-CAP-EXT Methodology"

Patias, N.; Rowe, F.; Cavazzi, S.; Arribas-Bel, D. "Sustainable urban development indicators in Great Britain from 2001 to 2016". *Landscape and Urban Planning*, 214, 104148. 10.1016/j.landurbplan.2021.104148

Liu, Y.; Singleton, A.; Arribas-Bel, D.; Chen, M. "Identifying and understanding road-constrained areas of interest (AOIs) through spatiotemporal taxi GPS data: A case study in New York City". *Computers, Environment and Urban Systems*, 86. 10.1016/j.compenvurbsys.2020.101592

Calafiore, A.; Palmer, G.; Comber, S.; Arribas-Bel, D.; Singleton, A. "A geographic data science framework for the functional and contextual analysis of human dynamics within global cities". *Computers, Environment and Urban Systems*, 85. 10.1016/j.compenvurbsys.2020.101539

2020 Comber, S.; Arribas-Bel, D.; Singleton, A.; Dolega, L. "Using Convolutional Autoencoders to Extract Visual Features of Leisure and Retail Environments". *Landscape and Urban Planning*, 202

Thompson, R. N.; Hollingsworth, T. D.; Isham, V.; Arribas-Bel, D.; Ashby, B.; Britton, T.; Challenor, P.; Chappell, L. H. K.; Clapham, H.; Cunniffe, N. J.; Dawid, A. P.; Donnelly, C. A.; Eggo, R. M.; Funk, S.; Gilbert, N.; Glendinning, P.; Gog, J. R.; Hart, W. S.; Heesterbeek, H.; House, T.; Keeling, M.; Kiss, I. Z.; Kretzschmar, M. E.; Lloyd, A. L.; McBryde, E. S.; McCaw, J. M.; McKinley, T. J.; Miller, J. C.; Morris, M.; O'Neill, P. D.; Parag, K. V.; Pearson, C. A. B.; Pellis, L.; Pulliam, J. R. C.; Ross, J. V.; Tomba, G. S.; Silverman, B. W.; Struchiner, C. J.; Tildesley, M. J.; Trapman, P.; Webb, C. R.; Mollison, D.; Restif, O. "Key ques-

tions for modelling COVID-19 exit strategies". *Proceedings of the Royal Society B: Biological Sciences*, 287(1932), 20201405. 10.1098/rspb.2020.1405

Puente-Ajovín, M.; Ramos, A.; Sanz-Gracia, F.; Arribas-Bel, D. "How sensitive is city size distribution to the definition of city? The case of Spain". *Economics Letters*, 197. 10.1016/j.econlet.2020.109643

Franklin, R.; Houlden, V.; Robinson, C.; Arribas-Bel, D.; Delmelle, E. C.; Demšar, U.; Miller, H. J.; O'Sullivan, D. "Who Counts? Gender, Gatekeeping, and Quantitative Human Geography". *The Professional Geographer*

Chen, M.; Fahrner, D.; Arribas-Bel, D.; Rowe, F. "A reproducible notebook to acquire, process and analyse satellite imagery". *REGION*, 7(2), R15-R46. 10.18335/region.v7i2.295

Olson, A. W.; Zhang, K.; Calderon-Figueroa, F.; Yakubov, R.; Sanner, S.; Silver, D.; Arribas-Bel, D. "Classification and Regression via Integer Optimization for Neighborhood Change". *Geographical Analysis*

Liu, Y.; Singleton, A.; Arribas-Bel, D. "Considering Context and Dynamics: A Classification of Transit-Oriented Development for New York City". *Journal of Transport Geography*, 85, 102711. 10.1016/j.jtrangeo.2020.102711

Chen, M.; Arribas-Bel, D.; Singleton, A. "Quantifying characteristics of the local urban environment through geotagged Flickr photographs and image recognition". *International Journal of Geo-Information*, 9, 264. 10.3390/ijgi9040264

Lumnitz, S.; Arribas-Bel, D.; Cortes, R. X.; Gaboardi, J. D.; Griess, V.; Kang, W.; Oshan, T. M.; Wolf, L.; Rey, S. "splot - visual analytics for spatial statistics". *Journal of Open Source Software*, 5(47), 1882. 10.21105/joss.01882

2019 Arribas-Bel, D.; Garcia-Lopez, M.; Viladecans-Marsal, E. "Building(s and) cities: Delineating urban areas with a machine learning algorithm". *Journal of Urban Economics*. 10.1016/j.jue.2019.103217

Comber, S.; Arribas-Bel, D.; Singleton, A.; Dong, G.; Dolega, L. "Building Hierarchies of Retail Centers Using Bayesian Multilevel Models". *Annals of the American Association of Geographers*. 10.1080/24694452.2019.1667219

Singleton, A.; Arribas-Bel, D. "Geographic Data Science". *Geographic Analysis*. 10.1111/gean.12194

Stubbings, P.; Peskett, J.; Rowe, F.; Arribas-Bel, D. "A Hierarchical Urban Forest Index Using Street-Level Imagery and Deep Learning". *Remote Sensing*, 11(12). 10.3390/rs11121395

Arribas-Bel, D. "A course on Geographic Data Science". *The Journal of Open Source Education*, 2(14), 42. 10.21105/jose.00042

Comber, S.; Arribas-Bel, D. "Machine learning innovations in address matching: A practical comparison of word2vec and CRFs". *Transactions in GIS*, 23(2), 334-348

Liu, Y.; Singleton, A.; Arribas-Bel, D. "A principal component analysis (PCA)-based framework for automated variable selection in geodemographic classification". *Geo-spatial Information Science*. 10.1080/10095020.2019.1621549

Davies, A. E.; Dolega, L.; Arribas-Bel, D. "Buy online collect in-store: Exploring grocery click & collect using a national case study". *International Journal of Retail and Distribution Management*

Chen, M.; Arribas-Bel, D.; Singleton, A. "Understanding the dynamics of urban areas of interest through volunteered geographic information". *Journal of Geographical Systems*

Dong, G.; Alexiou, A.; Arribas-Bel, D.; Wolf, L. "Inferring neighbourhood quality with property transaction records by using a locally adaptive spatial multi-level model". *Computers, Environment and Urban Systems*

2018 Arribas-Bel, D.; Reades, J. "Geography and Computers: Past, present, and future". *Geography Compass*

Arribas-Bel, D.; Bakens, J. "Use and Validation of Location-Based Services in Urban Research: An Example with Dutch Restaurants". *Urban Studies*, 56(5), 868:884

Arribas-Bel, D.; Bakens, J. "Spatial dynamics of cultural diversity in the Netherlands". *Environment and Planning B - Urban Analytics and City Science*. 10.1177/2399808318783748

2017 Arribas-Bel, D.; Tranos, E. "Characterizing the Spatial Structure(s) of Cities 'on the fly': the Space-Time Calendar". *Geographical Analysis*, 50(2), 162-181. 10.1111/gean.12137

Gerritse, M.; Arribas-Bel, D. "Concrete agglomeration benefits: do roads improve urban connections or just attract more people?". *Regional Studies*, 52(8), 1134-1149

Wolf, L.; Anselin, L.; Arribas-Bel, D. "Stochastic Efficiency of Bayesian MCMC in Spatial Econometric Models: An Empirical Comparison". *Geographical Analysis*, 50, 97-119. 10.1111/gean.12135

Arribas-Bel, D.; Patino, J.; Duque, J. "Remote sensing-based measurement of Living Environment Deprivation: Improving classical approaches with machine learning". *PLOS ONE*, 12(5), e0176684

Comber, S.; Arribas-Bel, D. "'Waiting on the train': The anticipatory (causal) effects of Crossrail in Ealing". *Journal of Transport Geography*, 64, 13-22

2016 Arribas-Bel, D.; Nijkamp, P.; Poot, J. "How diverse can measures of segregation be? Results from Monte Carlo simulations of an agent-based model". *Environment and Planning A*, 48(10), 2046-2066

Folch, D.; Arribas-Bel, D.; Koschinsky, K.; Spielman, S. "Spatial Variation in the Quality of American Community Survey Estimates". *Demography*, 53(5), 1535-1554

2015 Arribas-Bel, D.; Kourtit, K.; Nijkamp, P. "Cultural Diversity and Urban Buzz". *Environment and Planning C -- Government and Policy*, 33, 1-17

Arribas-Bel, D.; Gerritse, M. "From manufacturing belt, to rust belt, to college country - A Visual Narrative of the US Urban Growth". *Environment and Planning A*, 47(6), 1241-1253

Arribas-Bel, D.; Ramos, A.; Sanz Gracia, F. "The size distribution of employment centers within the U.S. Metropolitan Areas". *Environment and Planning B: Planning and Design*, 42, 23-39

Arribas-Bel, D.; Kourtit, K.; Nijkamp, P.; Steenbruggen, J. "Cyber Cities: Social Media as a Tool for Understanding Cities". *Applied Spatial Analysis and Policy*, 8(3), 231-247

2014 Arribas-Bel, D.; Sanz Gracia, F. "The Validity of the Monocentric City Model in a Polycentric Age: US Metropolitan areas in 1990, 2000 and 2010". *Urban Geography*, 35(7), 980-997

Arribas-Bel, D. "Accidental, Open and Everywhere: Emerging Data Sources for the Understanding of Cities". *Applied Geography*, 49, 45-53

Kourtit, K.; Nijkamp, P.; Arribas-Bel, D. "Migrant Entrepreneurs as Urban 'Health Angels' -- Contrasts in Growth Strategies". *International Planning Studies*

2013 Arribas-Bel, D.; Schmidt, C. R. "Self-organizing maps and the US urban spatial structure". *Environment and Planning B: Planning and Design*, 2, 362-371

Arribas-Bel, D.; Kourtit, K.; Nijkamp, P. "Benchmarking of World cities through self-organizing maps". *Cities*, 31, 248-257

Anselin, L.; Arribas-Bel, D. "Spatial fixed effects and spatial dependence". *Papers in Regional Science*, 92(1), 3-17

Amaral, P. V.; Anselin, L.; Arribas-Bel, D. "Testing for spatial error dependence in probit models". *Letters in Spatial Resource and Sciences*, 6(2), 91-101

2012 Arribas-Bel, D.; Koschinsky, J.; Amaral, P. V. "Improving the Multi-Dimensional Comparison of Simulation Results: A Spatial Visualization Approach". *Letters in Spatial and Resource Sciences*, 5(2), 55-63

Kourtit, K.; Nijkamp, P.; Arribas-Bel, D. "Smart Cities Perspective - a Comparative European Study by Means of Self-Organizing Maps". *Innovation: The European Journal of Social Science Research*, 25(2), 229-246

Arribas-Bel, D.; Sanz-Gracia, F.; Ximenez-de-Embún, D. P. "Kangaroos, cities and space: a first approach to the Australian urban system"

Kourtit, K.; Arribas-Bel, D.; Nijkamp, P. "High Performers in Complex Spatial Systems: A Self-Organizing Mapping Approach with Reference to The Netherlands". *The Annals of Regional Science*, 48(2), 501-527

2011 Arribas-Bel, D.; Nijkamp, P.; Schoelten, H. "Multidimensional urban sprawl in Europe: a self-organizing map approach". *Computers, Environment and Urban Systems*, 35(4), 265-275

Rey, S.; Anselin, L.; Folch, D. C.; Arribas-Bel, D.; Sastré Gutiérrez, M. L.; Interlante, L. "Measuring Spatial Dynamics in Metropolitan Areas". *Economic Development Quarterly*, 25(1), 54-64

Conference Proceedings

2026 Wagg, D.; Malleson, N.; Beltran, A.; Tipuric, M.; Arribas-Bel, D. "Using Earth Observation Informed Agent-Based Models to build a Scenario Planning Digital Twin for Local Energy Policies". `10.5194/egusphere-egu26-22325`

2021 Alharbi, Y.; Coenen, F.; Arribas-Bel, D. "Sustainable Development Goals Monitoring and Forecasting using Time Series Analysis"

2020 Alharbi, Y.; Coenen, F.; Arribas-Bel, D. "Sustainable Development Goal Relational Modelling: Introducing the SDG-CAP Methodology"

Alharbi, Y.; Arribas-Bel, D.; Coenen, F. "Sustainable Development Goal Relational Modelling: Introducing the SDG-RMF Methodology"

2019 Alharbi, Y.; Arribas-Bel, D.; Coenen, F. "Sustainable Development Goal Attainment Prediction: A Hierarchical Framework using Time Series Modelling"

Other academic articles

2026 Arribas-Bel, D.; Franklin, R. "Satellite Salvation? Why satellite imagery is (part of) the answer to our urban data needs". *Environment and Planning B: Urban Analytics and City Science*, 53(5). 10.1177/23998083261456945

Beecham, R.; Arribas-Bel, D. "Open spatial science as routine practice: Urban Data/Code contributions from GISRUK 2024". *Environment and Planning B: Urban Analytics and City Science*, 53(3). 10.1177/23998083261431404

2025 Arribas-Bel, D.; Metzler, B.; Lynn, S. "AI in land use". *Environment and Planning B: Urban Analytics and City Science*, 52(2). 10.1177/23998083251316384

2024 Malleson, N.; Franklin, R.; Arribas-Bel, D.; Cheng, T.; Birkin, M. "Digital twins on trial: Can they actually solve wicked societal problems and change the world for better?". *Environment and Planning B: Urban Analytics and City Science*, 51(6), 1181-1186. 10.1177/23998083241262893

Arribas-Bel, D.; Fleischmann, M. "In praise of (spatial) bundles". *Environment and Planning B: Urban Analytics and City Science*. 10.1177/23998083231224151

2021 Arribas-Bel, D.; Alvanides, S.; Batty, M.; Crooks, A.; See, L.; Wolf, L. "Urban data/code: A new EP-B section". *Environment and Planning B - Urban Analytics and City Science*, 48(9), 2517-2519. 10.1177/23998083211059670

2020 Rowe, F.; Maier, G.; Arribas-Bel, D.; Rey, S. "The Potential of Notebooks for Scientific Publication, Reproducibility and Dissemination". *REGION*, 7(3), E1-E5. 10.18335/region.v7i3.357

2018 Arribas-Bel, D.; Tranos, E. "Big Urban Data: Challenges and Opportunities for Geographical Analysis". *Geographical Analysis*. 10.1111/gean.12157

2017 Harris, R.; O'Sullivan, D.; Gahegan, M.; Charlton, M.; Comber, L.; Longley, P.; Brunsdon, C.; Malleson, N.; Heppenstall, A.; Singleton, A.; Arribas-Bel, D.; Evans, A. "More bark than bytes? Reflections on 21+ years of geocomputation". *Environment and Planning B: Urban Analytics and City Science*, 44(4), 598-617

2016 Arribas-Bel, D. "London: The Information Capital: 100 Maps and Graphics That Will Change How You View the City. James Cheshire and Oliver Uberti. London: Particular Books (Penguin), 2014.". *Papers in Regional Science*, 96(1), 223-224

Arribas-Bel, D. "Geocomputation: A Practical Primer, edited by Chris Brunson and Alex Singleton.". *Journal of Regional Science*, 56(1), 176-178

2015 Arribas-Bel, D. "Review of Simulation of Complex Systems in GIS, by Patrice Laglois". *Environment and Planning B: Planning and Design*, 42, 564-567

Arribas-Bel, D. "The spoken postcodes". *Regional Studies, Regional Science*, 2(1), 457-460

Arribas-Bel, D.; Graaff, T. "WooW-II: Workshop on open workflows". *REGION*, 2(2), R1-R2

2014 Arribas-Bel, D. "Spatial Data, Analysis, and Regression - A mini course". *REGION*, 1(1), R1

2012 Arribas-Bel, D.; Gerritse, M. "Monocentricity? Commuting flows visually.". *Featured Graphic, Environment and Planning A*, 44(9), 2041-2042

Book chapters

2024 Arribas-Bel, D.; Graser, A. "Geographic data science: a manifesto"

2023 Arribas-Bel, D.; Rowe, F.; Chen, M.; Comber, S. "An image library: The potential of imagery in (quantitative) social sciences". 10.4337/9781789903942

2020 Boeing, G.; Arribas-Bel, D. "GIS and Computational Notebooks". *The Geographic Information Science & Technology Body of Knowledge (1st Quarter 2021 Edition)*. 10.22224/gistbok/2021.1.2

2018 Arribas-Bel, D. "Statistics, Modelling, and Data Science". *Digital Geographies*, 129-140. SAGE

de Graaff, T.; Arribas-Bel, D.; Ozgen, C. "Demographic Aging and Employment Dynamics in German Regions: Modeling Regional Heterogeneity". *Modelling Aging and Migration Effects on Spatial Labor Markets*, 211-231. Springer, Cham

2017 Arribas-Bel, D.; de Graaff, T.; Rey, S. "Looking at John Snow's cholera map from the XXIst century: a practical primer on Reproducibility and Open Science". *Regional Research Fontiers*

2013 Kourtit, K.; Nijkamp, P.; Arribas-Bel, D. "The Creative Urban Diaspora Economy: A Disparity Analysis among Migrant Entrepreneurs.". *Handbook of Research Methods and Applications in Urban Economies*

Other

2025 Arribas-Bel, D. "El papel de la geografía en la revolución de datos: ejemplos desde la ciencia de datos geográficos". *Geografía, cambio global y vulnerabilidad territorial. Herramientas para el conocimiento de un mundo en transformación permanente*. Universidad de La Laguna (Spain). http://riull.ull.es/xmlui/handle/915/42264

2024 The Alan Turing Institute "Geospatial AI for Land Use, by The Alan Turing Institute". https://www.gov.uk/government/publications/geospatial-ai-for-land-use-by-the-alan-turing-institute/geospatial-ai-for-land-use-by-the-alan-turing-institute

2021 Darlington-Pollock, F.; Arribas-Bel, D.; Rowe, F. "What do 'left behind' areas looklike over time?". https://localtrust.org.uk

2019 Arribas-Bel, D. "Imagery: A better 'picture' of the city". https://cp.catapult.org.uk/2019/12/13/imagery-a-better-picture-of-the-city/

Rowe, F.; Patias, N.; Arribas-Bel, D. "Policy Brief: Neighbourhood Change and Trajectories of Inequality in Britain, 1971-2011". *UK2070 Commission*

Arribas-Bel, D. "Six 15-minute walks capture the diversity, history of Houston neighborhoods". *Urban Edge Blog*. Kinder Institute for Urban Research, Rice University. https://kinder.rice.edu/urbanedge/2019/04/08/maps-six-15-minute-walks-capture-diversity-history-houston-neighborhoods

2015 Arribas-Bel, D. "Does Twitter make us view cities differently?". *University of Birmingham Tumblr blog. May 15th, 2015*. http://unibirmingham.tumblr.com/post/119013810163/does-twitter-make-us-view-cities-differently

2014 Arribas-Bel, D. "Open Workflows for Open Regional Science". *NARSC Newsletter. Issue 1, No.2, June*. http://www.narsc.org/newsite/wp-content/uploads/2014/06/NARSCNewsJune2014-Final2.pdf

2013 Arribas-Bel, D. "Cultural diversity and increased urban buzz". *Migration: New Developments. Spring 2013*. NORFACE Research Programme on Migration. http://www.norface-migration.org/files/NORFACE_2013_single.pdf

Working Papers

2013 Arribas-Bel, D.; Florax, R. "Heterogeneity and Dependence in Geo-Referenced Micro Data: Random Effects vs True Contagion"

Arribas-Bel, D.; Sanz Gracia, F. "On the existence of agglomeration shadows at the metropolitan level"

2012 Anselin, L.; Amaral, P. V.; Arribas-Bel, D. "Technical Aspects of Implementing GMM Estimation of the Spatial Error Model in PySAL and GeoDaSpace". https://geodacenter.asu.edu/drupal_files/sperrorgmm_wp2.pdf

Research Visits

2023 CEMFI. Madrid (Spain). External Faculty.

2022 CEMFI. Madrid (Spain). External Faculty.

2021 CEMFI. Madrid (Spain). External Faculty.

2019 Research in Spatial Economics (Rise) group. Universidad EAFIT. Medellin (Colombia). Visitor.

Instituto Nacional de Pesquisas Espaciais (INPE, National Institute for Space Research). São José dos Campos (Brazil). Visitor.

Instituto de Pesquisa Econômica Aplicada (IPEA, Institute of Applied Economic Research). Brasilia (Brazil). Visitor.

Centro de Desenvolvimento e Planejamento Regional da UFMG (CEDEPLAR, Centre for Development and Regional Planning). Universidade Federal de Minas Gerais. Belo Horizonte (Brazil). Visitor.

2018 Center for Geospatial Sciences, University of California, Riverside. Riverside (USA). Visitor.

2017 Center for Spatial Data Science, University of Chicago. Chicago (USA). Visitor.

Spatial Structures in the Social Sciences, Brown University. Providence (USA). Visitor.

2015 Kinder Institute for Urban Research, Rice University. Houston (USA). July-August. Visiting scholar.

2014 Department of Economics, University of Houston. Houston (USA). June-August. Visiting scholar.

2011 Department of Spatial Economics, The Faculty of Economics and Business Administration, Vrije Universiteit. Amsterdam (The Netherlands). July-August. Visiting researcher.

2010 Department of Spatial Economics, The Faculty of Economics and Business Administration, Vrije Universiteit. Amsterdam (The Netherlands). September. Visiting PhD student.

GeoDa Center for Geospatial Analysis and Computation, Arizona State University. Tempe, AZ (USA). March-May. Visiting PhD student.

2009 Department of Spatial Economics, The Faculty of Economics and Business Administration, Vrije Universiteit. Amsterdam (The Netherlands). September-December. Visiting PhD student.

2008-09 GeoDa Center for Geospatial Analysis and Computation, Arizona State University. Tempe, AZ (USA). August-June. Visiting PhD student.

Invited Lectures

Keynote speeches

2026 "Understanding Cities with New Forms of Data". SPACERAISE summer school, Grand Sasso Science Institute. L'Aquila (Italy)

2022 "Making it easy for Regional Scientists to consider (urban) form & function". Spatial Economic Analysis Lecture. Regional Science Association International, British and Irish Section. July. Stirling (UK)

2021 "El papel de la Geografía en la revolución de datos: ejemplos desde la Ciencia de Datos Geográficos". XXVII Congreso de la Asociación Española de Geografía. December. Tenerife (Spain)

"Building 'Geographic Data Science…'". International Conference on Geospatial Information Sciences. November. Mexico

"Building 'Geographic Data Science…'". Geoinformation Science Symposium (virtual). October. Indonesia

Seminars

2026 "Beyond the Thesis: Helping your research find its way into the world". Smart Data Research UK. Knowledge Basecamp masterclass. Online. June

"Embeddings for the rest of us". Universidad de Zaragoza. June

"Making satellites more useful, usable and used in social research and policy". Universidad de Zaragoza. June

"Making satellites more useful, usable and used in social research and policy". University of Leeds. June

"How satellite imagery can support constituency-level policymaking". House of Commons Library. Online. June

2025 "The case for satellite imagery in social science and policymaking". Chinese Academy of Sciences. Beijing (China). May

"The case for satellite imagery in social science and policymaking". Southwest University. Chongqing (China). May

2023 "Decoding urban form and function in Great Britain". IOER Forum. Leibniz Institute of Ecological Urban and Regional Development. November

"Decoding urban form and function in Great Britain". Economic Geography and Spatial Economics Seminar Series. London School of Economics. October

"Urban Grammar - Decoding the socio-economic landscape with satellite imagery". Workshop on Migration, Climate Change and Economic Development. Universidad de Zaragoza. September

"Decoding the Socio-Economic landscape with satellite imagery". Bank of England. July

"Learning from Deep Learning - Lessons from using computer vision to identify (urban) form and function in open data satellite imagery". AI and Sustainable Cities workshop. The Alan Turing Institute, UCL & QMUL workshop. June

"Measuring (even) better the Urban-Rural continuum". JRC Webinar Series. Future of Cities. Online. April

"Developing detailed, consistent and scalable characterisations of urban form & function". Masters Gathering. Industrial Technology Research Institute, Taiwan (online). April

2022 "Making it easy for regional scientists to consider (urban) form & function". Economic Data Science Seminar Series. Office for National Statistics. November

"Open by Default - Developing reproducible, computational research". Royal Statistical Society, Merseyside chapter. Session on open data sources. June

"Urban Analytics@Turing". Brown bag series. Mansueto Institute for Urban Innovation. University of Chicago. May

2021 "Building 'Geographic Data Science...'". Diplomado en Análisis de Datos para las Ciencias Sociales (remote). Universidad de Caldas. Colombia

"Geographic Data Science: why, what, how". HM Land Registry Data Science seminars

"(Urban) Form & Function in Great Britain". Urban Analytics monthly meetup (remote). Alan Turing Institute (UK)

"Infusing our understanding of cities with Geographic Data Science". (remote). ETH Zurich (Switzerland)

"Spatial Signatures -- Understanding (urban) spaces through form and function". (remote). ESRC Urban Big Data Centre seminar series. University of Glasgow. Glasgow (UK)

2020 "Building 'Geographic Data Science...'". MSc in Economics seminar, Universidad de San Andrés (Argentina). Online. November

"Urban Analytics - Maximising the value of big data for enhanced urban policy making". Building Smart Cities with Citizens and for the Public Good: A Symposium. Organised by the Heseltine Institute, University of Liverpool. Liverpool. March

2019 "Using New Forms of Data & Machine Learning to Understand Cities". Innovation LAB seminar series, OECD. Paris (France). December

"Images as (policy-ready) data". Acto Anual de Política Regional y Fondos Europeos en España (Annual Event for Regional Policy and European Funds in Spain). Madrid (Spain). November

"Building(s and) Cities". MacArthur Foundation workshop on Urban Modelling & Complexity Science. Centre for Advanced Spatial Analysis, University College London. London (UK). September

"Infusing Urban Analytics with Geographic Data Science". Urban Analytics at the interface between environment, human health and behaviour workshop. Tongji University (China). July

"Urban Analysis with Geographic Data Science and New Forms of Data". Instituto Nacional de Pesquisas Espaciais (INPE) (Brazil). March

"Infusing Urban and Regional Policy with Geographic Data Science". Instituto de Pesquisa Economica Aplicada (IPEA) (Brazil). March

"Infusing Urban and Regional analysis with Geographic Data Science". CUSP Seminars, King's College London. London (UK). January

"Accessing & Visualizing Satellite Data". Workshop on Ocean and Coastal Monitoring and Modelling for Coastal Zone Management. Commonweath Marine Economies Programme. St. Vincent & Grenada. January

2018 "Infusing Urban and Regional analysis with Geographic Data Science". University of Toronto, Toronto (Canada). November

"Infusing Urban & Regional work with New Forms of Data". OECD, Paris (France). September

"Geographic Data Science - What, why, and how". 300,000Km/s, Barcelona (Spain). March

"An Updatable Index of Neighborhood Desirability". IEB Seminars, Universitat de Barcelona (Spain). February

"New methods for new data? A few urban examples". Data Science Seminars. ShopDirect, Liverpool. January

2017 "Towards a Geographic Data Science that Works for Everyone". Spatial Data Science Conference. CARTO HQ, Brooklyn. December

"The Spatial Structure of Cities Through The Lens Of New Forms Of Data". CASA Seminars. Centre for Advanced Spatial Analysis. University College London. November

"Looking at Cities with New Lenses". Roxby Centenary Lecture, University of Liverpool. November

"Bringing new forms of data to the study of cities". UK Data Service Webinar. February

"Remote Sensing-Based Measurement of Living Environment Deprivation - Improving Classical Approaches with Machine Learning". GVC/City REDI Seminar Series, University of Birmingham. (Birmingham, UK). February

"Bringing New Forms of Data to the Study of Cities". Ritsumeikan University (Kyoto, Japan). January

2016 "Smart Cities - Examples, Opportunities and Challenges from Urban Research". Smart Cities: myth or reality? event held by the Telefonica Chair on Smart Cities and the IEB. University of Barcelona (Spain). January

2015 "The (geo)data revolution and the need for (Geographic) Data Science". Remote lecture for the postgraduate course Research Design in Urban Science, coordinated by Professor Marina Alberti. University of Washington (US). October

"New approaches to measure the spatial structure(s) of cities". Geospatial Big Data seminars. University College London (UK). March

2014 "Seeing cities through the lens of new (big) data -- Some examples". University of Liverpool (UK). December

"Seeing cities through the lens of new (big) data". Institute for Employment Research (IAB). Nuremberg (Germany). September

"Spatial Analysis & Spatial Regression". Third year module in the Bachelor in Economics' Minor in Geography. Two lectures (3h). University of Utrecht (Netherlands). February

"New data & old Regional Science -- Some examples". GeoDa Center Brownbag series, Arizona State University (USA). February

"Agglomeration Lights and Shadows within metropolitan areas -- US Cities in 2000". Department of Global Economics & Management internal seminars, University of Groningen (Netherlands). February

2012 "GeoDaSpace: advanced spatial econometrics made easy". V Spatial Econometrics Seminar in honour of Jean Paelink, Universidade de Coimbra (Portugal). October

"An introduction to R (with Nick Malizia)". Graduates for Integrative Society and Environment Research (GISER), Arizona State University. March

2011 "Studying urban polycentricity in the US with spatial analysis methods". European Investment Bank. July

2007 "45 minutes on Geographical and Urban Economics". Bremen International Graduate School of Social Sciences. December

Conferences and Events

Organization

2025 *The Interesting Data and Methods Sessions*. Series of sessions organised with Rachel Franklin, Elizabeth Delmelle, and Isabelle Nilsson at the AAG Annual Meeting. Detroit (US). April

2023 *The Future of Urban Analytics*. Series of sessions organised between the Alan Turing Institute, the University of Liverpool, and the University of Leeds at the AAG Annual Meeting

2022 *Workshop: Urban Analytics in Policymaking*. OECD Laboratory for Geospatial Analysis. Online

2021-2023 *Spatial Analytics and Data - The Interviews*. Sponsored by the Royal Geographical Society and The Alan Turing Institute. With Rachel Franklin and Levi Wolf

2018 *The Potential of Satellite Imagery for Studying Human Activity*. workshop held at the University of Liverpool. Co-organised with Francisco Rowe. June

2017 *Geographic Data Science*. special sessions and panel for the Annual Meeting of the AAG , held in Boston, MA. Co-organized with Mark Birkin, Marynia Kolak, Nick Malleson, Alex Singleton and Seth Spielman

2016 *Big Data for Urban and Regional Analysis*. special sessions for the Annual Meeting of the AAG , held in San Francisco, CA. Co-organized with Emmanouil Tranos

2014/15 *Big Data Lunch Seminars*. co-organized with Emmanouil Tranos and University of Birmingham Institute of Advanced Studies

2015 *Big Data for Urban and Regional Analysis*. special sessions for the Annual Meeting of the AAG , held in Chicago, IL. Co-organized with David Folch and Emmanouil Tranos

2014 *Applications of new sources of (big) data in Regional Science*. special sessions for the 61st NARSC Meetings, held in Washington, DC. Co-organized with David Folch

Big Data Day. co-organized with Emmanouil Tranos and University of Birmingham Institute of Advanced Studies. September

Participation

2025 Annual Meeting of the AAG, Detroit (USA); GISRUK, Bristol (UK); Living Planet Symposium, Vienna (Austria)

2023 Annual Meeting of the AAG, Denver (USA)

2021 International Seminar on Urban Form. Glasgow (UK, online)

2018 Annual Meeting of the AAG, New Orleans (USA)

2017 Annual Meeting of the AAG, Boston (USA); GIS Research UK, GISRUK, Manchester (UK)

2016 Annual Meeting of the AAG, San Francisco (USA); Workshop on Big Data and Transport organized by the Joint Research Centre of the European Comission, Sevilla (Spain); Convening on Urban Data Science - University of Chicago, Chicago (USA)

2015 NARSC Meetings, Portland (OR, USA); GIS Research UK - GISRUK, Leeds (UK); Annual Meeting of the AAG, Chicago (USA)

2014 Western Regional Science Association Meeting, San Diego (USA); Tinbergen Institute Workshop, Amsterdam (NL); Workshop on Big Data and Urban Informatics, University of Illinois at Chicago (USA); Department of Economics, University of Houston (USA), NARSC Meetings

2013 Western Regional Science Association Meeting, Santa Barbara (USA); NORFACE Migration Conference, London (UK); Tinbergen Institute Workshop, Amsterdam (NL); ERSA Congress, Palermo (Italy); Dutch-Israeli Workshop on Regional-Science, Groningen (NL); 5th. NORFACE Migration Conference, Berlin (Germany); NARSC Meetings, Atlanta (USA)

2012 Spatial Econometrics Association World Conference, Salvador (Brazil); ERSA Congress, Bratislava (Slovakia); NORFACE "Increasing Heterogeneity in the Workforce and its Impact", Nuremberg (Germany)

2011 Eureka Seminar, VU University, Amsterdam (NL); ERSA Congress, Barcelona (Spain); NARSC Meetings, Miami (USA); Tinbergen Institute Complex-City Workshop, Amsterdam (NL)

2010 Workshop on Urban Dynamics URBANICS, Marbella (Chile); ERSA Congress, Jonköping (Sweden); Workshop Urban Systems 2.0., Delft (NL); FOSS4G, Barcelona (Spain)

2009 Spatial Econometrics Association World Conference, Barcelona (Spain); NARSC Meetings, San Francisco (USA)

2008 RESER, Sttutgart (Germany); Spatial Econometrics Association World Conference, Brooklyn (USA)

Teaching

Postgraduate

Geographic Data Science (2015-2020). https://darribas.org/gds_course

Web Mapping and Analysis (2020-). https://darribas.org/wmn

Spatial Modelling for Data Scientists (2021). https://gdsl-ul.github.io/san

Data Science Studio (2018-). https://darribas.org/data_science_studio

Spatial Analysis (2015-20). https://darribas.org/spa_notes

Undergraduate

Geographic Data Science (2015-). https://darribas.org/gds_course

Virtual Field Class (2021-)

Field Class Year 3, Barcelona (2015-20)

Geography undergraduate dissertation supervisions (2014-2020)

Field Class Year 1, Shropshire (2014)

Geography Year 1 and 2 Tutorials (2014-2020)

Research Methods for Dissertations (2014)

Workshops and Short Courses

2026 "Embeddings for the rest of us". SPACERAISE summer school, Grand Sasso Science Institute. L'Aquila (Italy)

"Embeddings for the rest of us". 4h workshop, jointly taught with Elisabetta Pietrostefani and Fang Chen. University of Birmingham (UK). GISRUK 2026. April

2023 "Geographic Data Science for Applied Economists". (remote). 15h course. CEMFI. Madrid (Spain). November

"Urban Analytics". (remote). 4h session at the MSc Engineers for Smart Cities. Université Côte D'Azur. November

"Spatial autocorrelation". Spatial Data Science Bootcamp (with Levi Wolf). CARTO. London. February

2022 "Super-Charging your (Regression) Model with Space and Geography". Together with Pedro Amaral. Center for Spatial Data Science. University of Chicago. May

2021 "Python for Geographic Data Science". Urban Systems Lab. Imperial College London. 2-day course

"Geographic Data Science for Regional Economists". (remote). 8h course. Universitat de València. Valencia (Spain)

"Geographic Data Science for Applied Economists". (remote). 15h course. CEMFI. Madrid (Spain)

2020 "Enriching your Analytics with Geographic Data Science". (remote). 2h workshop. Spatial Data Science Conference (online). CARTO

"Geographic Data Science for Regional Scientists". (remote). 2h workshop. Spanish Association of Regional Science (online) Conference

"Data Science studio in Python". (remote). 8h workshop. CDRC. UK

"Data Science studio in Python". (remote). 8h workshop. Universitat de Barcelona. Barcelona (Spain)

2019 "Introduction to machine learning in Python". One-day workshop. OECD. Paris (France)

"R & Py Spatial Analysis workshop". One-day workshop, jointly taught with Roger Bivand. European Colloquium on Theoretical and Quantitative Geography. Luxemourg

2018 "Geographic Data Science in Python". Three-day course. University of Toronto (Toronto). November

"Geographic Data Science in Python". Three-day course. University of Luxembourg (Luxembourg). September

"Introduction to Python for Data Science". Two-day course. OECD (Paris). September

"Python for Geographic Data Science". Two-day course for the Q-Step Programme. University College London (UK). June

"Introduction to Python for Geographic Data Science". One-day course for the Consumer Data Research Centre. University of Leeds (UK). May

2017 "Geographic Data Science for Urban Analytics". 4 day workshop jointly taught with Alex Singleton at the Ritsumeikan University. Kyoto (Japan). January

2016 "Geographic Data Science with PySAL and the pydata stack". 4h workshop jointly taught with Sergio J. Rey at the 15th. annual Scientific Computing with Python Conference. Austin (USA). July

2014 "Spatial Data, Analysis and Regression mini-course". 14h workshop taught at the Institute for Employment Research (IAB). Nuremberg (Germany). September

"Workshop on open Workflows II". 1.5 day workshop jointly taught with Thomas de Graaff at the Department of Spatial Economics, VU University. Amsterdam (Netherlands). September

"Spatial Data, Analysis and Regression mini-course". 12h workshop taught at the Department of Economics, University of Houston. Houston (USA). August

2013 "Geo-computation in Python". Workshop taught at the VII Jornadas de SIG Libre. March 6th. SIGTE, Universitat de Girona (Spain). March

2012 "Modeling Spatial Systems: Networks and Spatial Econometrics". One day intensive course jointly taught with Dr. Emmanouil Tranos. Department of Spatial Economics, VU University. Amsterdam (Netherlands). December

"An introduction to spatial analysis and spatial econometrics with OpenGeoDa and GeoDaSpace". Salvador (Brazil). July

"OpenGeoDa, GeoDaSpace and PySAL Lab instructor". Spatial Econometrics Advanced Institute. Catholic University. Rome (Italy). May

2009 "Spatial Data Analysis in Ecology and Conservation". hosted by the National Centre for Biological Sciences in collaboration with the Nature Conservation Foundation and the GeoDa Center. Bangalore (India). July

Assistantship

2010 "Microeconomics II". Bachelor in Business Administration, Department of Economic Analysis, Universidad de Zaragoza (Spain). Principal instructor: Gemma Larramona. September-December

"Mejorando la experiencia de aprendizaje de los estudiantes de microeconomía intermedia. Elaboración de cuestionarios de autoevaluación en Moodle con feedback automático". (Project "Improving the learning experience of intermediate microeconomics students. Fabrication of self-evaluation

questionaires in Moodle with automatic feedback"). PIIDUZ-10-2-492. Universidad de Zaragoza. Member. 01/09/2010-30/06/2011. Coordinator: Domingo Pérez Ximénez de Embún

Line management - Supervision

Postdoctoral Researchers

2025- Martina Pardy. PDRA, Imago Data Service for Imagery

Fang Chen. PDRA, Imago Data Service for Imagery

Shaonlee Patranabis. PDRA, Imago Data Service for Imagery

2023-25 Stuart Lynn. Lead Research Data Scientist

Barbara Metzler. PDRA

2020-24 Martin Fleischmann. PDRA

2022-23 Anna Zanchetta. PDRA

2021-22 Alessia Calafiore. PDRA

Supervisor

2025- Baromasak Klanreungsang. University of Liverpool (Primary supervisor)

2024- Yuta Sato. ESRC NWSSDTP

2020-2025 Gladys Kenyon. ESRC CDAS CDT, with Idealista (Primary supervisor)

2020-24 Qihao Liu. University of Liverpool

2019-25 James Murphy. ESRC CDAS CDT, with Marlan Maritime Technologies (Primary supervisor)

2018-24 Danial Owen. ESRC CDAS CDT (Primary supervisor)

2018-23 Sian Teesdale. ESRC CDAS CDT, with ESRI-Here. Unfinished

2022 John Freeman. ESRC CDAS CDT, with Health & Safety Executive. Unfinished (Primary supervisor)

Meruyert Kenzhebay. ESRC CDAS CDT, with MGISS. Unfinished

2017-22 Melanie Green. ESRC CDAS CDT, with Ordnance Survey. Unfinished (Primary supervisor)

2017-23 Krasen Samardzhiev. ESRC CDAS CDT, with CARTO (Primary supervisor)

2018-22 Yassir Alharbi. University of Liverpool

2017-21 Nikos Patias. ESRC CDAS CDT, with Ordnance Survey

Sam Comber. ESRC NWSSDTP, with Local Data Company (Primary supervisor)

2016-21 Meixu Chen. University of Liverpool (Primary supervisor)

Yunzhe Liu. University of Liverpool

Ellen Talbot. ESRC NWSSDTP

2016-18 Muhammad Hakim Bin Danial. University of Liverpool

Kush Tackar. University of Liverpool

Visitor Host

2024-25 Paula Falcao. Universidade Federal de Minas Gerais. Brazil

2023-24 Ugur Ursavas. Zonguldak Bülent Ecevit University. Turkey

2022 Marianna Magagnoli. Universitat de Barcelona. Spain

2019 Júlio Cesar Lopes. Universidade Católica de Brasília. Brazil

2017 Cassiano Ricardo Dalberto. Universidade Federal de Minas Gerais. Brazil

Committee Member

2025 Job Rosier. Characterization of urban land use and urban land dynamics in data scarce regions. VU University (Amsterdam). External examiner. April

2021 Mohammed Ibrahim. A Computer Vision System for Detecting and Analysing Critical Events in Cities. University College London. External examiner. April

Sarah Catherine Gadd. Analysing spatio-temporal networks with continuous-time models. University of Leeds. External examiner. April

2020 Michele Ferretti. Geographies of Code: Decompiling the 'Smart City' through the study of software engineering practices. Kings College London (UK). External examiner. December

Jacob Macdonald. Leveraging Geospatial Statistics for Measuring and Valuing the Urban Environment. Universidade Nova de Lisboa (Portugal). Committee member. July

Helena Nilsson. Spatial Organisation of Retail Activities. Jönköping University. Examiner. May

2019 Alec Edward Davies. The era of repurposed data: Exploring health via Data Science and new forms of data. University of Liverpool. Internal examiner. January

2018 Helen V. Roberts. Investigating the role of social media and smart device applications in understanding human-environment relationships in urban green spaces. University of Birmingham. External examiner. February

2016 Linda Woods. A Statistical and Geographical Analysis of Workplace Accidents in Great Britain. University of Liverpool. Internal examiner. December

2015 Jorge Eduardo Patiño Quinchía. Citiscape, Poverty and Crime: a Quantitative Assessment Using VHR Imagery. Universitat Politècnica de València. External referee and committee member. December

Master

2012 Hada Melissa Sáenz Vela. "Análisis exploratorio de Características Socioeconómicas de México mediante la aplicación de métodos de agregación espacial, 2010". Universidad Autónoma de Cohauila, Centro de Investigaciones Socioeconómicas. Lector (member of Committee). September

Service to University and Community

Journal Referee

African Journal of Agricultural Research (1), Annals of the American Association of Geographers (1), Annals of GIS (1), Annals of Regional Science (4), Applied Geography (3), Applied Spatial Analysis and Policy (1), Computers, Environment and Urban Systems (12), Environment and Planning A (2) / B (12) / C (1), European Planning Studies (1), European Urban and Regional Studies (1), Geographical Analysis (8), Innovation (1), International Journal of Sustainable Development (1), International Journal of GIS (2), International Regional Science Review (6), Journal of Geographical Systems (1), Journal of Geography and Regional Planning (1), Journal of Housing and the Built Environment (1), Journal of Regional Science (1), Journal of Statistical Software (2), Journal of Transport Geography (2), Journal of Urban Management (1), Journal of Urbanism (1), Landscape and Urban Planning (1), Nature Human Behaviour (1), Papers in Regional Science (5), Proceedings of the Python in Science Conference (2), REGION (1), Regional Science Policy and Practice (1), Regional Studies (4), Spatial Economic Analysis (4), Spatial Networks (1), Tourism Economics (1), Transactions in GIS (3), Urban Geography (2). Urban Studies (3)

External Examination Experience

2024- University College London (UK): MSc in Social and Geographic Data Science

2022-2025 University of Leeds (UK): MSc in Urban Data Science and Analytics

2019-2023 University of Glasgow (UK): MSc in Urban Analytics

External Review/Panel Experience

2021 UKRI. Agile Research and Innovation response to COVID-19 issues (EPSRC Panel). Panel member

2019 Wellcome. Open Research Fund Awards. Panel member

2018 Jisc Geographic Data Service. Test & Feedback Panel. Member

Other

2023- Co-host, The GLaD Podcast. https://gladpodcast.podbean.com

2024 Scientific Committee Member, World Multidisciplinary Congress on Civil Engineering, Architecture, and Urban Planning. Ostrava (Czech Republic)

2018 Mentor, Google Summer of Code. Project: "splot: visualisation in PySAL"

2017 Mentor, Google Summer of Code. Project: "Regionalization in PySAL"

2016 Mentor, Google Summer of Code. Project: "Spatial Interaction Modeling in PySAL"

Paper reviewer of the proceedings, 2016 Python for Scientific Computing (SciPy) Conference

2015 Paper reviewer of the proceedings, 2015 Python for Scientific Computing (SciPy) Conference

2014- Young Scholars Committee Member. Centre for Geocomputation. Ryerson University (Canada)

Paper reviewer of the proceedings, 2014 Python for Scientific Computing (SciPy) Conference

Program committee member, 2014 Python for Scientific Computing (SciPy) Conference

2013 Epainos Award / Young Regional Scientist Sessions. Reviewer. 53rd. ERSA Congress

Scientific Software

PySAL — Python library of spatial analysis. Core development team. https://pysal.org

gds_env — A containerised platform for Geographic Data Science. Creator and maintainer. https://darribas.org/gds_env/

contextily — Contextual tiles in Python. Creator and core developer. https://contextily.readthedocs.io/

xyzservices — XYZ basemaps in Python. Creator and developer. https://xyzservices.readthedocs.io/

GeoDaSpace — Advanced spatial econometrics. Core development team.

Language Skills

Spanish — Native.

English — Fluent writing, reading and speaking. Maximum level by the Institute of Modern Languages at the University of Zaragoza.

French — Medium writing, reading and speaking. Fourth year (out of five) at the Institute of Modern Languages at the University of Zaragoza.

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